Day 06: IPv4 Addressing

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Cybersecurity Learning Log

Part 1: IPv4 Theory

1. What is IPv4?
IPv4 stands for Internet Protocol version 4. It is a protocol used to address devices and route data packets across networks.

2. How many bits are in an IPv4 address?
An IPv4 address contains 32 bits.

3. What is an octet?
An octet is a group of eight bits. It can represent a decimal value from 0 to 255.

4. How many octets are in an IPv4 address?
There are four octets in an IPv4 address, usually written as four decimal numbers separated by dots.

5. What is a subnet mask?
A subnet mask identifies which bits of an IPv4 address belong to the network portion and which belong to the host portion.

6. Difference between network portion and host portion?
The network portion identifies the network, while the host portion identifies an interface within that network. The subnet mask determines how the address is divided.

7. What is a private IPv4 address?
A private IPv4 address is an address reserved for use within private networks, such as home, school, and office networks. These addresses are not directly routed across the public internet.

8. Write all three private IPv4 ranges.
10.0.0.0 - 10.255.255.255 (10.0.0.0/8)
172.16.0.0 - 172.31.255.255 (172.16.0.0/12)
192.168.0.0 - 192.168.255.255 (192.168.0.0/16)

9. What is a public IPv4 address?
A public IPv4 address is an address that can be routed on the public internet, subject to routing policies and allocation rules.

10. What is the IPv4 loopback address?
127.0.0.1 is the commonly used IPv4 loopback address. It allows a device to communicate with its own network stack.

11. What is APIPA?
APIPA stands for Automatic Private IP Addressing. On Windows, it can automatically assign an address in the 169.254.0.0/16 range when a device cannot obtain an IPv4 address from DHCP.

12. What does 169.254.x.x commonly indicate?
It commonly indicates that a device has assigned itself a link-local IPv4 address, often because it could not obtain an address from a DHCP server. It does not by itself prove the exact cause.

13. What is a default gateway?
A default gateway is the router or next-hop device a computer uses to send traffic to destinations for which it has no more specific route.

14. Static vs dynamic IP addressing?
A static IP address is configured manually or reserved to remain consistent. A dynamic IP address is assigned automatically, usually by DHCP, and may change over time.

15. What is DHCP?

DHCP stands for Dynamic Host Configuration Protocol. It automatically provides network configuration, such as an IP address, subnet mask, default gateway, and DNS server.

16. What is a network address?

A network address identifies a specific IP network. Its host bits are all set to zero for the relevant subnet.

17. What is a broadcast address?

An IPv4 subnet broadcast address is used to send a packet to all hosts on that subnet. In a conventional subnet, its host bits are all set to one.

18. Why is IPv4 knowledge important in cybersecurity?

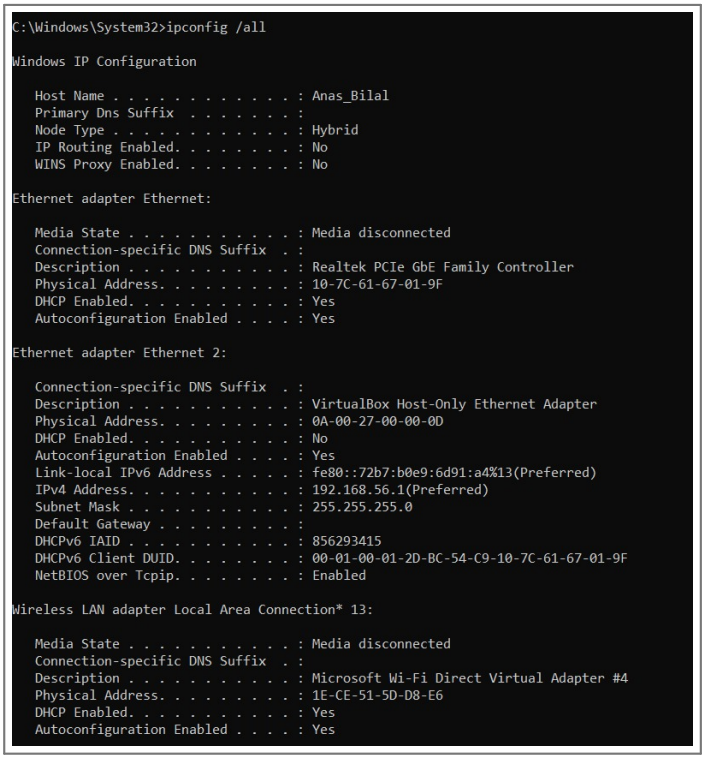
IPv4 knowledge helps security professionals understand network traffic, identify source and destination addresses, recognize private and public addressing, troubleshoot connectivity, investigate suspicious connections, and understand network segmentation.

Part 2: Practical Identification

Item	Value	Source
IPv4 address	192.168.1.38	Figure 2 (ipconfig /all)
Subnet mask	255.255.255.0	Figure 2 (ipconfig /all)
Default gateway	192.168.1.1	Figure 2 (ipconfig /all)
DNS server	8.8.8.8 (primary), 8.8.4.4 (secondary)	Figure 2 (ipconfig /all)
DHCP enabled	Yes (DHCP server 192.168.1.1)	Figure 2 (ipconfig /all)

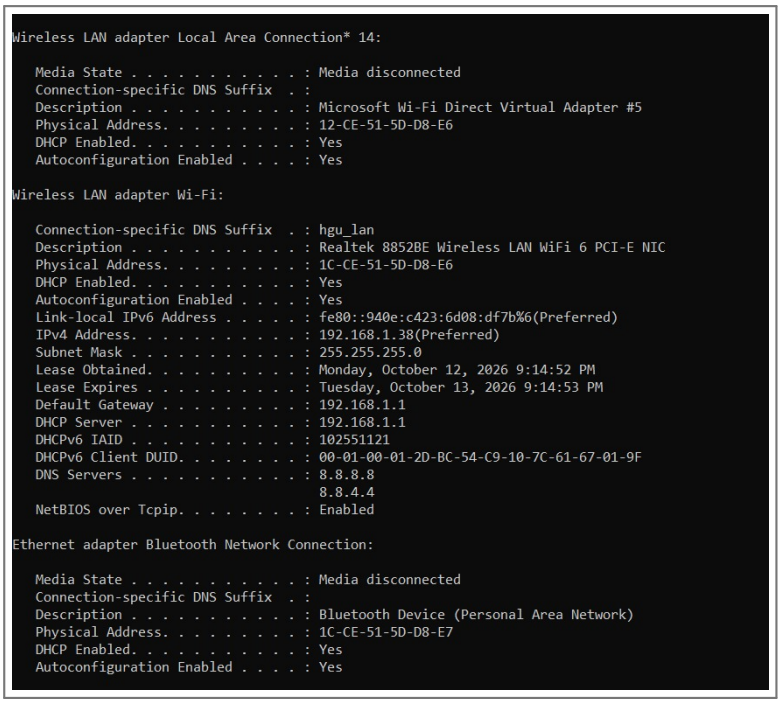
Part 3: Screenshots

Figure 1: ipconfig /all - Host Details and Ethernet Adapters



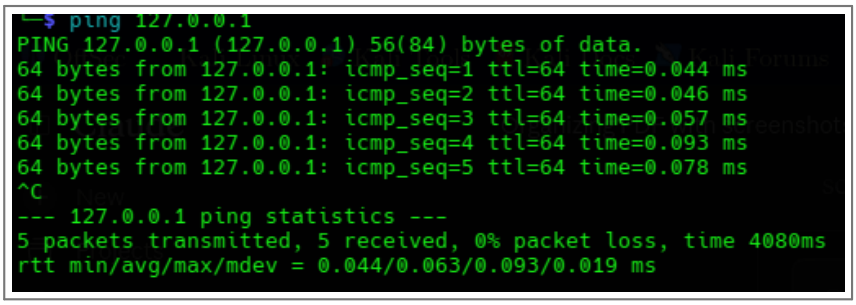
The **Ethernet** adapter shows *Media disconnected*. **Ethernet 2** is a VirtualBox Host-Only adapter with the IPv4 address **192.168.56.1**, subnet mask 255.255.255.0, DHCP disabled and no default gateway.

Figure 2: ipconfig /all - Active Wi-Fi Adapter



The active **Wi-Fi** adapter has DHCP enabled and received the IPv4 address **192.168.1.38** with subnet mask **255.255.255.0**. The default gateway and DHCP server are **192.168.1.1**, and the DNS servers are **8.8.8.8** and **8.8.4.4**. The address is in the private 192.168.0.0/16 range.

Figure 3: Loopback Test (ping 127.0.0.1)



ping 127.0.0.1 sent 5 packets and received 5 (0% packet loss), with an average time of 0.063 ms and TTL 64. This confirms the local TCP/IP stack responds. It does not test internet connectivity.

Conclusion

IPv4 addressing helps devices communicate across local and remote networks. Understanding subnet masks, gateways, DHCP, and private address ranges provides an important foundation for network troubleshooting and cybersecurity investigations.